Thermodynamic description of the Bi–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Bi–Sn binary system as calculated with pycalphad from the SGTE dataset BiSn-08Li.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Bi) + (Sn), lies at 415.8 K.

1. Introduction

The Bi–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of Z. Li, Z. Cao, S. Knott, A. Mikula, Y. Du, Z. Qiao, Calphad, 32, 152-63(2008). The calculations use pycalphad 0.11.2 at 1 atm.

For Bi, the liquidus meets the axis at 544.5 K. The melting point of Sn reproduced by the dataset is 505.1 K.

2. Phase diagram

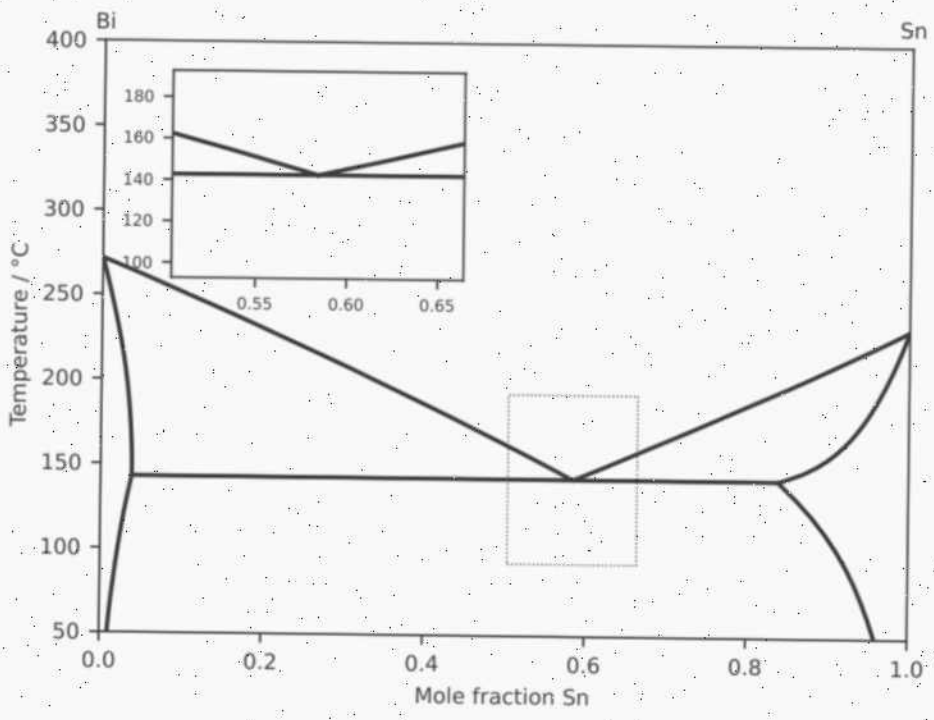


Fig. 1. Calculated Bi–Sn phase diagram (inset: enlarged view near the lowest invariant).

At 415.8 K the eutectic L → (Bi) + (Sn) takes place; the L phase then contains 58.4 at.%

Sn.

Table 1. Invariant reactions in the Bi–Sn system (calculated).

Reaction	Type	T (°C)	x(Sn), 1	x(Sn), 2	x(Sn), 3
L → (Bi) + (Sn)	eutectic	142.7	0.584	0.038	0.838

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 650 K the integral mixing enthalpy of the liquid has its maximum, 121 J/mol, at 64.0 at.% Sn.

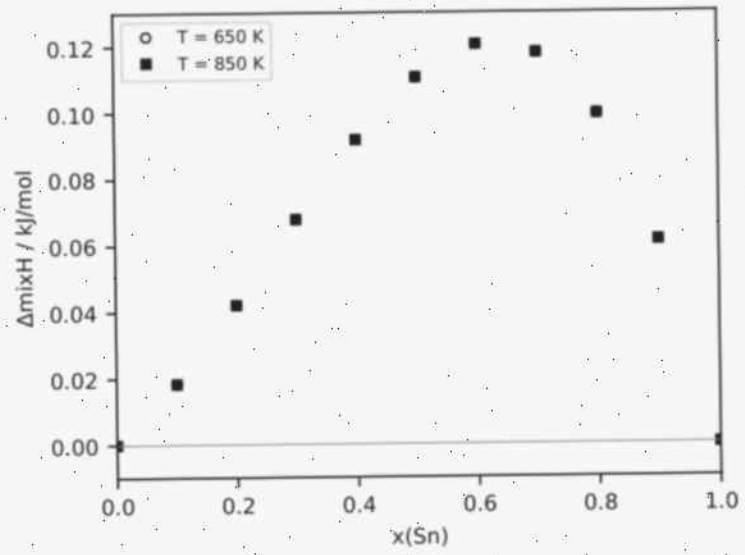


Fig. 2. Calculated mixing enthalpy of the liquid at 650 K and 850 K.

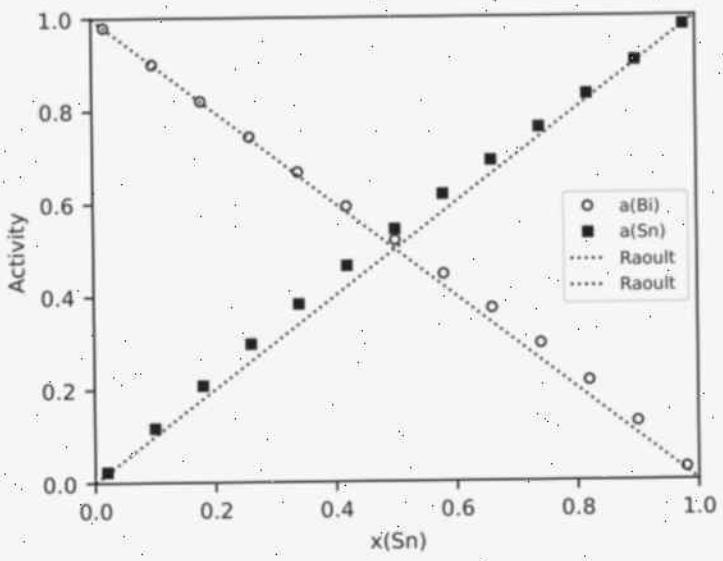


Fig. 3. Calculated activities in the liquid at 650 K (reference: pure liquids).

Table 2. Redlich–Kister parameters of the liquid (Bi–Sn).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	441.8	1.33493	0.0
L1	−298.1	−0.28229	0.0

Values as printed in the TDB file.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.